

## PAPER

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[View Journal](#) | [View Issue](#)Cite this: *Catal. Sci. Technol.*, 2018, 8, 1113Enhanced Fischer–Tropsch performances of graphene oxide-supported iron catalysts *via* argon pretreatment†Yuxue Wei,<sup>ac</sup> Chenghua Zhang,<sup>id</sup> \*<sup>ab</sup> Xi Liu,<sup>ab</sup> Yi Wang,<sup>b</sup> Qiang Chang,<sup>ac</sup> Ming Qing,<sup>b</sup> Xiaodong Wen,<sup>id</sup> \*<sup>ab</sup> Yong Yang<sup>ab</sup> and Yongwang Li<sup>\*ab</sup>

Graphene oxide (GO)-supported iron oxide nanoparticles (NPs) have been successfully synthesized by a hydrothermal method and used as Fischer–Tropsch synthesis (FTS) catalysts. GO with oxygen-containing groups may have a strong interaction with iron oxide NPs, retarding the contact between active sites and syngas. Argon pretreatment is used to modify the physical and chemical properties of catalysts with the purpose of enhancing FTS performances. The catalysts are thoroughly characterized by N<sub>2</sub> physisorption, X-ray diffraction, X-ray photoelectron spectroscopy, Raman spectroscopy, scanning electron microscopy, transmission electron microscopy, temperature-programmed reduction, and Mössbauer spectroscopy. The pretreatment of catalysts in argon at 500 °C for 5 h can effectively increase the surface area and defects, weaken the Fe–GO interaction and expose more catalytically active sites. The Fe/GO-500 catalyst shows improved reduction and carburization behaviors, high CO conversion activity and C<sub>5+</sub> selectivity, and excellent stability in the FTS reaction.

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## 1. Introduction

Fischer–Tropsch synthesis (FTS) can convert syngas derived from coal, natural gas and biomass into liquid fuels. FTS has been widely applied in industry and extensively studied in academia. Iron-based catalysts are often used in FTS due to their low cost, wide temperature range, tunable product distribution and high water gas shift (WGS) activity.<sup>1</sup> It is well known, on alumina or silica supported catalysts, that iron species will form iron aluminates or silicates due to strong metal–support interactions.<sup>2</sup> These aluminates or silicates are difficult to reduce. Therefore, the formation of an active iron carbide phase may be hindered. To overcome the shortcomings of these supports, it is necessary to find a carrier material with limited interaction toward iron. In this sense, carbon materials can be ideal candidates for catalyst supports. They can not only provide vast surface areas for high dispersion of active metals but also exhibit moderate and tunable interactions with metals. A number of research studies have been reported in using carbon materials as supports for FTS cata-

lysts. Vannice *et al.*<sup>3</sup> impregnated iron oxide nanoparticles (NPs) on activated carbon for CO hydrogenation. The iron oxide crystallites on activated carbon show a high olefin/paraffin ratio and better stability. Wei Chen *et al.*<sup>4</sup> prepared CNT-encapsulated iron oxide particles and iron oxide particles dispersed on CNT outer walls for FTS. The iron oxide encapsulated in CNTs exists in a more reduced state and shows higher activity and C<sub>5+</sub> selectivity. Among carbon materials, graphene oxide (GO) has been given particular attention due to its large surface area, low cost and sufficient oxygen-containing groups. Moussa *et al.*<sup>5</sup> reported graphene-supported iron oxide NPs for FTS *via* a microwave irradiation-assisted reduction method using iron nitrates and hydrazine hydrate. Compared with CNTs, graphene-supported Fe–K NPs show high activity and C<sub>8+</sub> selectivity with excellent stability. This is attributed to the defects present in the graphene lattice which act as anchoring sites for NPs. Ding Ma and co-workers<sup>6</sup> employed an incipient wetness impregnation method to disperse iron oxide NPs onto graphene oxide. The effects of different H<sub>2</sub> treatment temperatures of GO on FTS performances are investigated. GO reduced in 873 K as a support shows significant activity and C<sub>5+</sub> selectivity. This is ascribed to the weaker interaction between iron oxide NPs and graphene due to the decrease of oxygen-containing groups. Recently, Baoning Zong and his group<sup>7</sup> prepared a reduced graphene oxide-supported K-promoted iron catalyst for Fischer–Tropsch synthesis for olefin. FeK1/rGO shows a maximum FTY of 646 μmol<sub>CO</sub> g<sub>Fe</sub><sup>−1</sup> s<sup>−1</sup> and FeK2/rGO presents the

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highest selectivity to lower olefin of 68%. The aforementioned research reports demonstrate the superior FTS performances of GO-supported iron catalysts. However, a study in Ding Ma's group<sup>6</sup> implies that the residual oxygen species in GO can suppress the FTS activity and reduce the chain growth probability. Thus, the GO is required to be either annealed in hydrogen at temperatures as high as 800 °C or reduced by hydrazine hydrate to partly remove the oxygen species to enhance the FTS activity. These pretreatment processes are complicated, expensive or rigorous for the preparation of iron-based FTS catalysts. From this point of view, it is of significance to find an easily-controlled pretreatment method for catalyst activation and obtain a deep understanding of the intrinsic effect of graphene oxide as a substrate for FTS, especially without reductants.

In this paper, thermal pretreatment in argon is used to modify the interaction between Fe and GO, in order to expose more catalytically active sites. The pretreated catalysts then show enhanced FTS activities. N<sub>2</sub> physisorption, Raman spectroscopy (Raman), X-ray diffraction (XRD), Mössbauer spectroscopy (MES), X-ray photoelectron spectroscopy (XPS), scanning electron microscopy (SEM), transmission electron microscopy (TEM) and H<sub>2</sub>/CO temperature-programmed reduction (H<sub>2</sub>/CO-TPR) are used to characterize the fresh, pretreated and spent catalysts.

## 2. Experimental section

### 2.1 Catalyst preparation

Aqueous graphene oxide sol (10 mg mL<sup>-1</sup>) was bought from Carmery Graphene Marketing Center, Institute of Coal Chemistry, Chinese Academy of Sciences. The Fe/GO composites were synthesized through a hydrothermal method by introducing Fe(NO<sub>3</sub>)<sub>3</sub> as the Fe<sup>3+</sup> provider into GO water solution.<sup>8</sup> Typically, 1.4 g Fe(NO<sub>3</sub>)<sub>3</sub> was added to 80 mL GO water solution with stirring. After stirring for 1 h, the homogeneous suspension was transferred to a Teflon-lined autoclave which was placed in an oven at 120 °C for 12 h. After cooling down to room temperature, the products were washed with ethanol several times and then dried at 80 °C, named Fe/GO.

The products were then pretreated in argon at 300, 350, 400, 450 and 500 °C for 5 h, respectively. They were denoted as Fe/GO-300, Fe/GO-350, Fe/GO-400, Fe/GO-450 and Fe/GO-500, respectively.

### 2.2 Catalyst characterization

The concentration of the metal in the catalyst was determined by inductively coupled plasma-optical emission spectrometry (ICP-OES, Thermo Icap 6300, USA). The morphology of the samples was determined by scanning electron microscopy (SEM) on FEI Quanta 400 FEG equipment. Transmission electron microscopy (TEM) was performed on an FEI Talos F200A electron microscope operating at an accelerating voltage of 200 kV. The samples were crushed and dispersed in ethanol under ultrasonic vibration and finally dispersed on a holey carbon film on a copper grid. The textural proper-

ties of the catalysts were determined *via* N<sub>2</sub> physisorption at -196 °C, using a Micromeritics ASAP 2420 instrument. Before the measurement, the samples were degassed under vacuum at 120 °C for 1 h and 200 °C for 6 h.

X-ray diffraction (XRD) patterns of the fresh and calcined catalysts were obtained using a D2/max-RA X-ray diffractometer (Bruker, Germany), with Cu K $\alpha$  radiation at 30 kV and 10 mA. X-ray photoelectron spectroscopy (XPS) was performed on a Thermo Scientific K-alpha using Al K $\alpha$  ( $h\nu$  = 1486.6 eV) as an X-ray source. The catalyst powders were dispersed onto conductive carbon tape that was pasted on the sample holder. The binding energy was calibrated using the adventitious carbon C 1s peak (284.8 eV).

The Raman spectra were recorded on a Lab RAM HR800 (Horiba Jobin-Yvon), using a charge-coupled device as a detector. A 532 nm air cooled frequency doubled Nd-Yag laser was employed as the excitation source with a power of 7 mW. The scanning time was set to 120 s. The samples (100–120 mesh) were supported by a glass slide, on which the Raman spectrum was collected at room temperature.

H<sub>2</sub>/CO-TPR was carried out using a dynamic analyzer (Micromeritics, Model 2920). About 50 mg of the sample was loaded into a U-type quartz tube reactor. The reactor was heated from room temperature to 850 °C at a heating rate of 10 °C min<sup>-1</sup>. The CO<sub>2</sub> (CO-TPR) signal was detected with an on-line quadrupole mass spectrometer.

The Mössbauer spectra of samples were acquired on an MR-351 constant-acceleration Mössbauer spectrometer (FAST, Germany) at 10 K. The radioactive source was a <sup>57</sup>Co source in a Rh matrix. The spectrometer was operated in a symmetric constant-acceleration mode. The spectra were collected over 512 channels in the mirror image format.

The XRD patterns of catalysts after the FTS reaction and the phase transformation of the catalysts during *in situ* reaction were determined with a Bruker D8 Advance XRD using Co K $\alpha$  radiation at 35 kV and 40 mA. The *in situ* reaction system was specially designed and supplied by UCT. The reaction was carried out in a borosilicate capillary (1.0 mm outside diameter, 0.02 mm wall thickness, 75 mm length). The catalysts were *in situ* reduced with a stream of H<sub>2</sub> at 300 °C for 6 h. The FTS reaction was carried out in a flow of syngas (H<sub>2</sub>/CO = 2) at 280 °C.

### 2.3 FTS test

The FTS reactions were carried out in a fixed bed reactor. In a typical process, 200 mg catalyst (40–60 mesh) was placed in the flat-temperature zone of the stainless steel tubular reactor to eliminate the temperature gradient. The remaining volume of the reactor was filled with quartz granules (40–60 mesh). For this granule size, it can be seen that the CO conversion becomes independent of the external mass transfer limitations.<sup>9</sup> All the tested samples were reduced in a stream of H<sub>2</sub> at 300 °C for 24 h. The FTS reaction was carried out in a flow of syngas (H<sub>2</sub>/CO = 2) at 280 °C, 2 MPa and a gas velocity of 10 L per g of catalyst per h. A detailed description of the

reactor and the product analysis system has been provided elsewhere.<sup>10</sup> The catalytic activity was expressed in terms of CO conversion and Fe time yield (FTY). The CO conversion was determined by comparing the reactant gas and product gas chromatograms. The FTY was defined as moles of CO per gram of iron per second. The hydrocarbon selectivities were calculated on a carbon basis, with the exception of CO<sub>2</sub>.

### 3. Results and discussion

#### 3.1 Textural properties

Table 1 lists the results of N<sub>2</sub> physisorption for fresh and pretreated catalysts. The N<sub>2</sub> adsorption and desorption isotherms are given in the ESI.† A high BET surface area of 192 m<sup>2</sup> g<sup>-1</sup> is found on the fresh Fe/GO sample. With the calcination in argon at 300 °C for 5 h, the BET surface area remarkably decreases from 192 m<sup>2</sup> g<sup>-1</sup> to 152 m<sup>2</sup> g<sup>-1</sup>, due to the decomposition of oxygen-containing groups under high temperature calcination conditions.<sup>11,12</sup> Some sintering of iron oxide NPs also contributes to the decrease in the surface area. However, further increasing the calcination temperature to 500 °C, the BET surface area increases to 162 m<sup>2</sup> g<sup>-1</sup>. Importantly, an apparent increment in the pore volume is observed from 0.11 cm<sup>3</sup> g<sup>-1</sup> for Fe/GO to 0.16 cm<sup>3</sup> g<sup>-1</sup> for Fe/GO-500. Considering that GO will be partly removed and even rolled during the calcination process, the observed increment in Fe/GO-500 surface area is primarily attributed to the voids and pores formed in the composite, which can provide more active sites to adsorb syngas.

#### 3.2 Morphology and phase

The morphologies of catalysts are examined by SEM and TEM, as shown in Fig. 1. The low-magnification SEM image (Fig. 1a) displays an overall perspective of the composites. The Fe/GO possesses a layer-by-layer assemble structure consisting of GO and iron oxide NPs. From the TEM images, it is clearly shown that the obtained iron oxide has a mixed rod- and particle-like morphology and homogeneously disperses on GO. More importantly, even after a long time of sonication in TEM sample preparation, the NPs are still firmly anchored on the surface of micrometer-scale GO, suggesting the strong interaction between the two substances.<sup>13</sup> The NPs have an average size of 39 nm measured by using Nano Measurer software (counting more than 200

nanoparticles, shown in the inset of Fig. 1b). In the HRTEM image (Fig. 1c), the presence of GO layers around iron oxide NPs can be clearly identified. This is probably due to the rolling and shrinking of GO sheets during the hydrothermal process. An interlayer distance of 0.27 nm in the core matches well with the (104) plane of  $\alpha$ -Fe<sub>2</sub>O<sub>3</sub> (JCPDS 89-8104). Few-layer graphene nanosheets with a lattice spacing of 0.34 nm are observed around the nanocrystals, implying that some  $\alpha$ -Fe<sub>2</sub>O<sub>3</sub> NPs may be covered with graphene sheets.<sup>14</sup> For Fe/GO-300, the iron oxide nanocrystals and GO sheets can be clearly observed from the SEM image (Fig. 1d), further confirming the GO layer-coated structure of the catalysts. The scanning transmission electron microscopy (STEM) images of Fe/GO-300 and the corresponding mapping are given in the ESI.† The mixed rod- and particle-like iron NPs of about 38.5 nm are still well dispersed on GO sheets as shown in Fig. 1e. Obvious lattice planes can be easily found in the HRTEM image (Fig. 1f) and the corresponding interplanar spacing (0.34 nm) matches with the (002) plane of graphene. With the increase of calcination temperature, the sizes of Fe/GO-350, -400 and -450 slightly increase to 39, 40.8 and 42.6 nm, respectively. The TEM images of them are presented in the ESI.† For Fe/GO-500, the perfect GO sheets are destroyed to some extent with more apertures and defects (Fig. 1g). The high temperature also promotes the aggregation of small NPs increasing the sizes to 45.2 nm. In addition, the argon pretreatment at 500 °C produces more open channels and creates large voids between GO sheets, allowing the inner surfaces of the composites to be effectively utilized. The formation of these defects is probably due to the removal of basal plane C–OH groups on GO,<sup>11</sup> leading to the breakage of the GO membrane-like structure. Thus, the BET surface area of Fe/GO-500 increases and more iron oxide NPs are exposed. From the STEM image and the corresponding Fe, O and C mappings in Fig. 1i, we can clearly see that a nanoparticle is partly exposed with some pores. In addition, there are more near-spherical iron oxide NPs formed, due to the aggregation of small nanorods or NPs under high temperature calcination. The HRTEM image of the NPs in the inset of Fig. 1i also confirms the existence of pores. Moreover, these pores are almost isolated from each other.

The XRD pattern (Fig. 2) of the fresh Fe/GO catalyst corresponds well to that of hematite ( $\alpha$ -Fe<sub>2</sub>O<sub>3</sub> JCPDS 89-8104), signifying the successful growth of  $\alpha$ -Fe<sub>2</sub>O<sub>3</sub> NPs on graphene under hydrothermal conditions. After calcination in argon, the XRD patterns of Fe/GO-300, Fe/GO-350 and Fe/GO-400 are similar to that of Fe/GO. On the basis of the Scherrer equation, the crystallite size of  $\alpha$ -Fe<sub>2</sub>O<sub>3</sub> NPs in Fe/GO-300 is estimated to be *ca.* 15.2 nm from the full width at half-maximum of the most intensive (104) peak. It is slightly smaller than that of Fe/GO (15.8 nm). The large difference in the measured sizes by XRD and TEM is attributed to the presence of some small nanorods/NPs. These small nanorods/NPs in the samples are difficult to count due to the coverage of GO sheets, resulting in the relatively large average crystallite sizes measured by TEM. Further increasing the

**Table 1** Textural properties of the as-prepared catalysts

| Catalysts | <i>S</i> <sub>BET</sub><br>(m <sup>2</sup> g <sup>-1</sup> ) | Average pore<br>size (nm) | Pore volume<br>(cm <sup>3</sup> g <sup>-1</sup> ) | Fe content <sup>a</sup><br>(wt%) |
|-----------|--------------------------------------------------------------|---------------------------|---------------------------------------------------|----------------------------------|
| Fe/GO     | 192                                                          | 3.1                       | 0.11                                              | 24.38                            |
| Fe/GO-300 | 152                                                          | 3.3                       | 0.12                                              | 31.88                            |
| Fe/GO-350 | 132                                                          | 3.6                       | 0.13                                              | 32.22                            |
| Fe/GO-400 | 123                                                          | 4.0                       | 0.14                                              | 32.88                            |
| Fe/GO-450 | 119                                                          | 3.6                       | 0.13                                              | 33.51                            |
| Fe/GO-500 | 162                                                          | 3.6                       | 0.16                                              | 34.87                            |

<sup>a</sup> Determined by ICP.



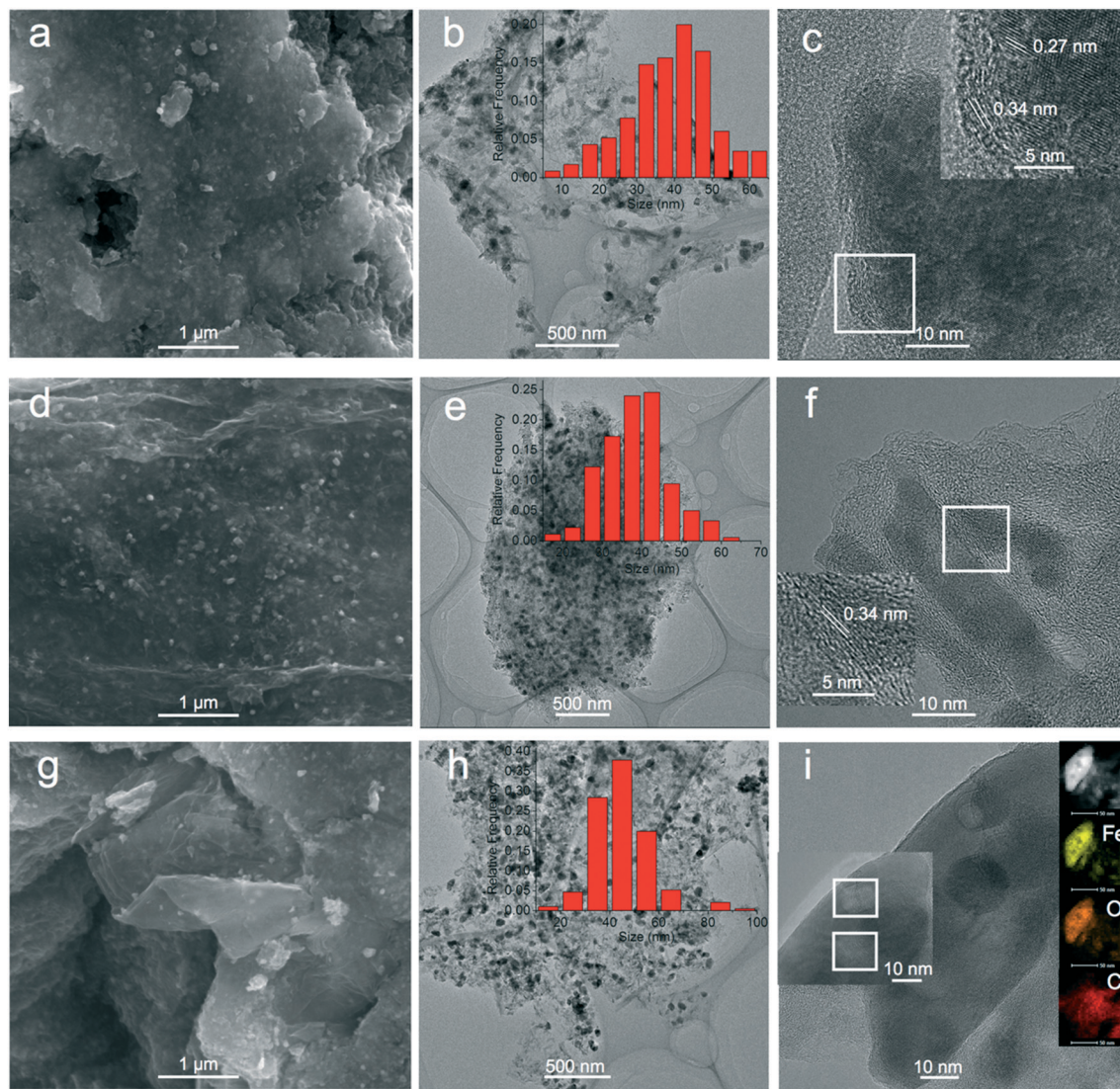


Fig. 1 SEM and TEM images of (a)–(c) Fe/GO; (d)–(f) Fe/GO-300; (g)–(i) Fe/GO-500.

calcination temperature to 350 °C, the crystallite size is 15.3 nm and is almost kept unchanged for Fe/GO-400. However, the XRD patterns of Fe/GO-450 and Fe/GO-500 match well with either the maghemite  $\gamma\text{-Fe}_2\text{O}_3$  (JCPDS 39-1346) or magnetite  $\text{Fe}_3\text{O}_4$  (JCPDS 19-629) phase. Since the XRD patterns of maghemite and magnetite are highly similar, Raman and XPS are used to identify the phase. Fig. 3 shows the Raman spectra of the composites. Three obvious peaks at 217, 280 and 391  $\text{cm}^{-1}$  are characteristic of  $\text{Fe}_2\text{O}_3$ .<sup>15</sup> It is reported by Han and co-workers that the typical Raman band of the  $\text{Fe}_3\text{O}_4$  phase is located at 660  $\text{cm}^{-1}$ .<sup>15</sup> The absence of the characteristic peak at 660  $\text{cm}^{-1}$  of Fe/GO-450 and Fe/GO-500 excludes the possibility of the  $\text{Fe}_3\text{O}_4$  phase. Fig. 4b presents the Fe 2p XPS spectra of the composites. The peaks at 710.9 eV and 724.5 eV are assigned to Fe 2p<sub>3/2</sub> and Fe 2p<sub>1/2</sub>, which match well with  $\text{Fe}_2\text{O}_3$ . The satellite peak with a binding energy of 718.5 eV is characteristic of  $\text{Fe}^{3+}$ , representing the existence of  $\text{Fe}_2\text{O}_3$  rather than  $\text{Fe}_3\text{O}_4$ . Therefore, combined with the XRD results, it is reasonable that the phase of Fe/GO-450

and Fe/GO-500 is maghemite  $\gamma\text{-Fe}_2\text{O}_3$ . No obvious diffraction peaks of  $\alpha\text{-Fe}_2\text{O}_3$  appear, indicating that the initial  $\alpha\text{-Fe}_2\text{O}_3$  phase completely transformed into  $\gamma\text{-Fe}_2\text{O}_3$  after calcination at 450 and 500 °C. It is widely accepted that the  $\alpha\text{-Fe}_2\text{O}_3$  phase is the most stable iron phase and less stable  $\gamma\text{-Fe}_2\text{O}_3$  usually transforms into  $\alpha\text{-Fe}_2\text{O}_3$  under calcination.<sup>16</sup> However, in this study, the phase transformation from  $\alpha\text{-Fe}_2\text{O}_3$  to  $\gamma\text{-Fe}_2\text{O}_3$  occurs at 450 °C in an argon atmosphere. Considering that the phase composition of calcination products depends on the temperature, time, pressure, rate of gas flow, chemical composition and structure of the original material and particularly particle sizes,<sup>16</sup> the unique structure of the as-prepared graphene-supported iron NPs may account for this unusual phase transformation. The special thermal, morphological and surface properties of graphene seem to significantly affect the phase of composites. Although the exact mechanism of phase transformation from  $\alpha\text{-Fe}_2\text{O}_3$  to  $\gamma\text{-Fe}_2\text{O}_3$  is unclear until now, research about this phase transformation and the effect of the initial  $\alpha\text{-Fe}_2\text{O}_3$  and  $\gamma\text{-Fe}_2\text{O}_3$  phases

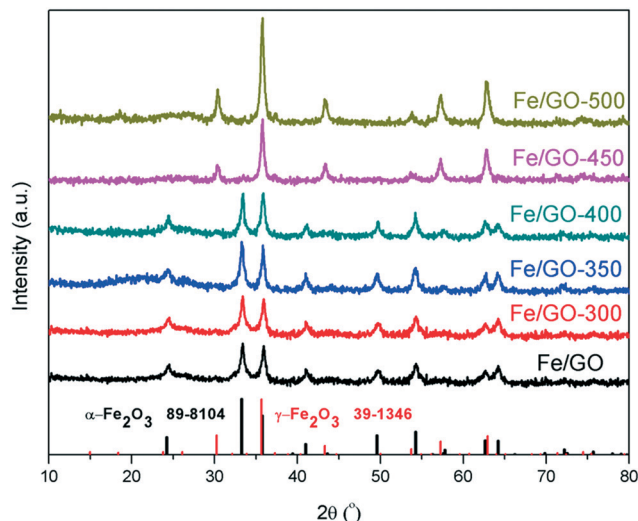


Fig. 2 XRD patterns of the catalysts.

on FTS performances is under progress. And we hope to resolve this matter in the near future. The XRD peaks are slightly narrower for Fe/GO-450 and Fe/GO-500, manifesting the increase in the sizes of iron oxide NPs. The crystallite sizes calculated from the (311) peak at  $35.6^\circ$  are *ca.* 18.8 and 19.2 nm for Fe/GO-450 and Fe/GO-500, respectively. Moreover, the high temperature calcination also promotes the crystallinity of iron oxide NPs. Both of the increased sizes and improved crystallinity account for the narrower XRD peaks in Fe/GO-450 and Fe/GO-500. No diffraction peaks at about  $25^\circ$  which is characteristic of graphene are detected, suggesting that the graphene sheets are disorderly stacked with a low degree of graphitization.<sup>17</sup>

### 3.3 Surface properties

To obtain more information on the nature of defects and disorders on the catalysts, Raman spectra are collected. Fig. 3 shows the Raman spectra of the fresh and calcined catalysts.

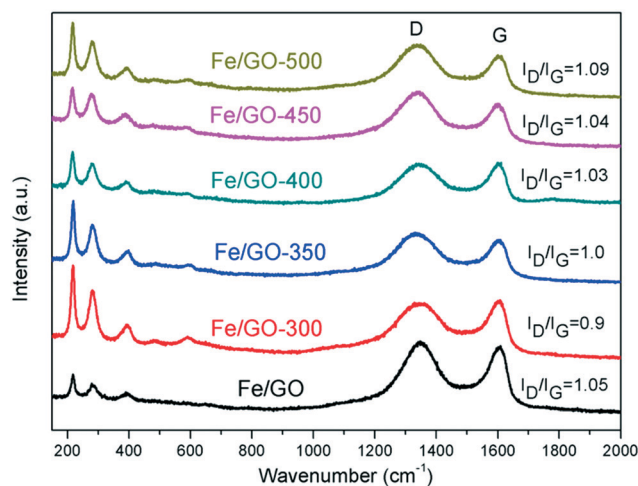


Fig. 3 Raman patterns of the catalysts.

Three obvious peaks at 217, 280 and  $391\text{ cm}^{-1}$  are characteristic of  $\text{Fe}_2\text{O}_3$ . After calcination, the characteristic bands at 217, 280, 391 and  $601\text{ cm}^{-1}$  corresponding to the  $\text{Fe}_2\text{O}_3$  phase are greatly enhanced. The strong peak at around  $1353\text{ cm}^{-1}$  is assigned to structural disorders and has been named band D. The other band at  $1603\text{ cm}^{-1}$  is ascribed to more graphitized carbon and is often called band G. The relatively weaker peak G in Fe/GO-500 indicates the low degree of graphitization.<sup>15</sup> The value of the D/G intensity ratio ( $I_D/I_G$ ) is used to evaluate the disorder degree in graphene sheets. The defects and disorders have certain advantages in the catalytic reaction. These defect sites provide nucleation and anchoring sites for iron NPs, reducing the mobility of particles and decreasing their tendency to agglomerate.<sup>5</sup> Chen *et al.* have reported that defect-rich carbon nanotube-supported platinum NPs have the highest hydrogen generation rate and durability.<sup>18</sup> Compared with Fe/GO, the  $I_D/I_G$  ratio for the Fe/GO-300 composite decreases from 1.05 to 0.9. Upon further increasing the calcination temperature to 450 and  $500^\circ\text{C}$ , the  $I_D/I_G$  ratio increases to 1.04 and 1.09 correspondingly, implying the presence of more structural defects in the composites. The damage and even loss of the membrane-like GO structure under high temperature calcination, as confirmed by BET and TEM measurements, enhance the structure disorders and defects within GO sheets. No other features corresponding to different iron phases are detected, supporting the above results that both the  $\text{Fe}_2\text{O}_3$  phase and GO exist in the composites.

The surface information of catalysts is obtained from XPS. The XPS survey spectra (Fig. 4a) show that the sample surface consists of Fe, C, and O elements. It is obvious that the Fe 2p, C 1s and O 1s peaks shift to lower binding energy compared with those of Fe/GO due to the modification of the Fe-GO interaction by argon pretreatment. Fig. 4b shows the Fe 2p XPS spectra of catalysts. By comparing the fresh Fe/GO catalyst with calcined catalysts, the binding energy of Fe 2p obviously shifts to lower values. The binding energy shift of Fe 2p indicates the electron transfer from GO to Fe species. The GO surface is in an electron-rich state, which easily transfers electrons to Fe species. This results in an electron-rich state of Fe species, thus the binding energy of Fe decreases. It is reasonable that the argon calcination modifies the interaction between GO and Fe species, promoting the electron transfer from GO to Fe. Therefore, Fe species easily form iron carbide species, which have been recognized as FT active phases.

For Fe/GO, the asymmetrical and broad features of the C 1s XPS peaks (Fig. 4c) suggest the co-existence of distinguishable models. Signal deconvolution with 80% Gaussian and 20% Lorentzian curve fitting<sup>11</sup> points out chemically different C species in the GO. The C 1s binding energies at about 284.8, 286.9 and  $288.5\text{ eV}$  are usually assigned to C-C/C=C, C-O (epoxy/alkoxy) and C=O,<sup>6</sup> respectively. The latter two are from oxygen-containing groups formed on the surfaces and edges of GO sheets. These oxygen-containing groups in GO provide active sites to connect to  $\text{Fe}^{3+}$  ions in  $\text{Fe}(\text{NO})_3$ .



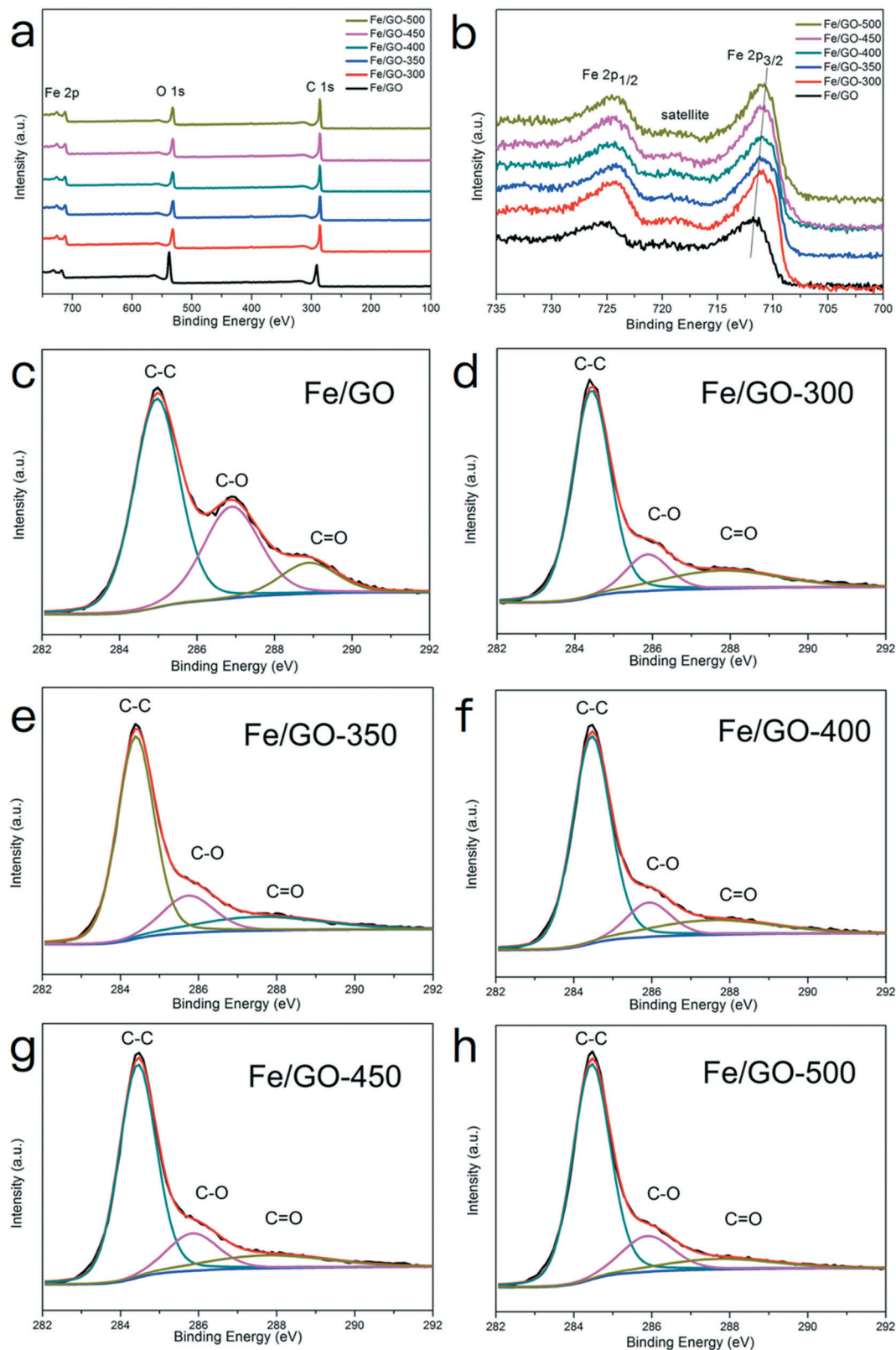


Fig. 4 XPS patterns of catalysts: (a) survey; (b) Fe 2p; (c)–(h) C 1s.

solution and guarantee the dispersion of iron oxide NPs. The summary of the area ratios of C–C/C=C, C–O and C=O is given in the ESI.† It is evident that a considerable decrease of

C–O and C=O is observed after calcination, which is consistent with literature reports.<sup>11,19</sup> Ma and co-workers have confirmed that less oxygen-containing groups in graphene lead

to a weaker interaction between GO and iron oxide.<sup>6</sup> Therefore, judging from the less oxygen-containing groups existing in the calcined catalysts, the argon pretreatments do weaken the Fe–GO interaction. However, there are slightly discernible differences in the content of oxygen-containing groups in different temperature calcined catalysts. Hence, it is justified to exclude the dominant impact of residual oxygen-containing groups on GO in calcined catalysts.

### 3.4 Reduction behavior

The reduction behaviors of the catalysts are measured by H<sub>2</sub>/CO-TPR. Fig. 5a shows the H<sub>2</sub>-TPR profiles of the catalysts. The Fe/GO catalyst without calcination shows a reduction peak of GO at 150 °C, indicating the reduction of oxygen-containing groups in GO sheets. After calcination in argon, only two well-separated H<sub>2</sub> consumption peaks can be detected, which are features of the two-step reduction of iron-based catalysts from Fe<sub>2</sub>O<sub>3</sub> to Fe<sub>3</sub>O<sub>4</sub> and Fe<sub>3</sub>O<sub>4</sub> to  $\alpha$ -Fe. It is obvious that the increment of calcination temperature shifts the position of reduction peaks to lower temperatures. The reduction temperature of Fe/GO-500 is similar to that of the Fe<sub>2</sub>O<sub>3</sub> reference (ESI<sup>†</sup>), suggesting that the argon pretreatment promotes the reduction of catalysts and the Fe–GO interaction in Fe/GO-500 is weakened. The breakage of the GO intact structure under high temperature calcination makes the access of hydrogen to iron NPs easier, thus low reduction temperature is achieved. The relatively increased size of iron NPs may also account for the lower reduction temperature.<sup>20</sup> In addition, the broad shoulder in the TPR profiles above 500 °C is likely due to the different degrees of interactions between the iron NPs and support that depend on size.<sup>21</sup> It is clear that the shoulder in Fe/GO-500 shifts to lower temperature compared with others, further confirming the weaker interaction between GO and iron oxide.

The CO-TPR (Fig. 5b) profiles also have two obvious peaks except in Fe/GO. The first peak located at 290–400 °C can be attributed to the reduction of Fe<sub>2</sub>O<sub>3</sub> to Fe<sub>3</sub>O<sub>4</sub>. Apparently, the temperature of this reduction peak decreases with the in-

crease of calcination temperature, which is consistent with H<sub>2</sub>-TPR. The second peak at 400–750 °C is assigned to the carburization of catalysts. The increase of calcination temperature presents obvious influence on the carburization peaks. Further increasing the calcination temperature to 500 °C significantly improves the carburization of the catalyst. The carburization peak of Fe/GO-500 becomes enlarged and ends in a temperature (*ca.* 710 °C) lower than for other catalysts (*ca.* 780 °C), which means easier carburization. This is in agreement with the observation in H<sub>2</sub>-TPR.

These results clearly demonstrate that the argon pretreatments promote the reduction of catalysts. Higher reducibility of Fe/GO-500 is observed by H<sub>2</sub>/CO-TPR. Several studies so far have reported that less oxygen-containing groups in graphene leads to a weaker interaction between iron oxide NPs and graphene,<sup>5–7</sup> which in turn enhances the reduction. However, the different temperature calcined catalysts have slightly discernible changes in the contents of oxygen-containing groups, as shown in the XPS results. In this case, at least three suppositions can be used to explain this phenomenon. First, the high temperature argon pretreatments produce open channels and create large voids between GO sheets, making more iron oxide NPs to be available for H<sub>2</sub> or CO reduction, as inferred from the BET, HRTEM and Raman results. The high accessible surface area and porosity in Fe/GO-500 promote continuous reduction of mass transport resistance to gas, thus easy reduction in H<sub>2</sub> or CO is achieved. Second, the interaction between larger iron oxide NPs and GO is much weaker than in smaller particles,<sup>20</sup> leading to the decreased reduction temperature with the increase of calcination temperature for the increased nanoparticle sizes, as confirmed by XRD and TEM. Third, the transfer of unpaired electrons on GO to iron oxide NPs is promoted by argon pretreatment, which is demonstrated by XPS. This renders the reduction of iron oxide easier and shifts the reduction to lower temperatures, coinciding with the autoreduction of iron oxide/CNT encapsulation.<sup>22,23</sup>

The easiest reduction and carburization behaviors of Fe/GO-500 in H<sub>2</sub> or CO play an important role in FTS. Since all

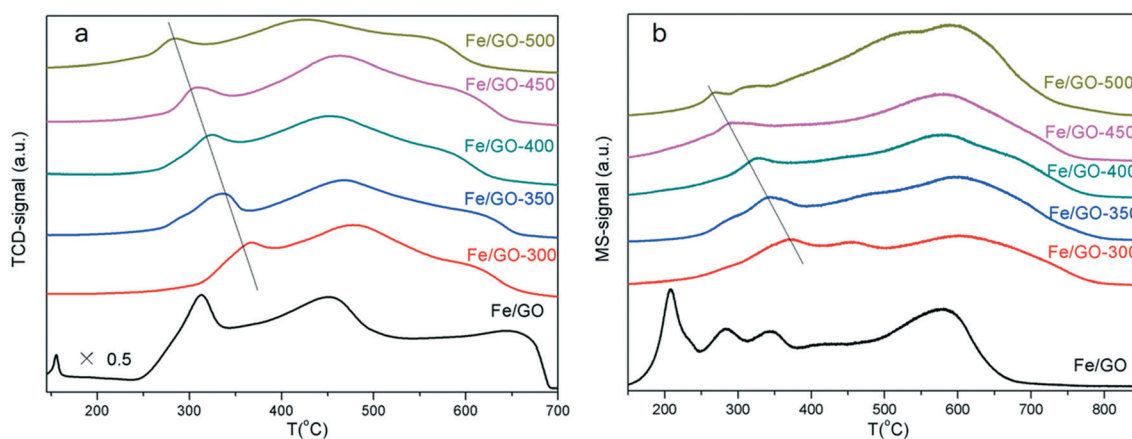


Fig. 5 (a) H<sub>2</sub>-TPR and (b) CO-TPR profiles of the catalysts.

catalysts are reduced at 300 °C before FTS, it is possible that the degree of reduction should be varied for the catalysts investigated. The extent of reduction does result in a difference in activity.<sup>20</sup> In summary, the characterization results reveal that the surface area, phase, morphology, surface properties and reduction behaviors of catalysts are strongly influenced by argon pretreatment with different temperatures.

### 3.5 FTS results

Table 2 summarizes the FTS results over catalysts under the reaction conditions of 280 °C, 2 MPa, 10 L per g of catalyst per h and H<sub>2</sub>/CO = 2. Since the catalysts have different iron contents, the FTY (Fe time yield) is used to compare the activity of catalysts. The FTY is defined as moles of CO per gram of iron per second. For the Fe/GO catalyst, the FTY is 30  $\mu\text{mol}_{\text{CO}} \text{g}_{\text{Fe}}^{-1} \text{s}^{-1}$ , and it is difficult to detect the product selectivity. Upon calcining the sample at 300 °C, the FTY significantly increases to 109  $\mu\text{mol}_{\text{CO}} \text{g}_{\text{Fe}}^{-1} \text{s}^{-1}$  and the heavy hydrocarbon (C<sub>5+</sub>) selectivity is 22%. The increment of calcination temperature leads to an increase in the activity, as shown by the increasing FTY from 109  $\mu\text{mol}_{\text{CO}} \text{g}_{\text{Fe}}^{-1} \text{s}^{-1}$  for Fe/GO-300 to 231.6  $\mu\text{mol}_{\text{CO}} \text{g}_{\text{Fe}}^{-1} \text{s}^{-1}$  for Fe/GO-500. The CO conversion of Fe/GO-450 and Fe/GO-500 is 22.2% and 39%, respectively. They are substantially higher than that of the Fe/PGO catalyst (10.2%) without H<sub>2</sub> pretreatment as reported by Ma and his group.<sup>6</sup> Another interesting result is the enhanced C<sub>5+</sub> selectivity over the 450 °C and 500 °C argon pretreated catalysts. The C<sub>5+</sub> selectivity increases from 22% for Fe/GO-300 to 37.7% for Fe/GO-500, in correspondence with the decrease of CH<sub>4</sub> selectivity from 29.5% to 15%. This is quite remarkable since graphene-supported iron catalysts without reductants (hydrogen or hydrazine hydrate) usually give low selectivity to C<sub>5+</sub>.<sup>6</sup> The C<sub>5+</sub> selectivity of Fe/GO-500 is relatively lower than that of potassium-promoted precipitated iron-manganese<sup>24</sup> and graphene-supported iron NPs Fe<sub>15</sub>K<sub>5</sub>-G.<sup>5</sup> This can be ascribed to the presence of a potassium promoter, which can suppress the H<sub>2</sub> chemisorption and thus shift the products to higher molecular weight.<sup>24</sup> It is worth mentioning that the olefin/paraffin ratios are very low on these catalysts, which are quite different from the results on other graphene-supported catalysts reported in the literature.<sup>7,25</sup> The differences can be attributed to the different electronic structures of graphene-supported catalysts. The catalyst surfaces in an

electron-rich state can suppress the hydrogenation of double bonds and improve the olefin selectivity.<sup>26</sup> In a study reported by Bao *et al.*,<sup>25</sup> N-doped graphene acts as an efficient electron donor to iron catalysts to enhance the selectivity for light olefins. The olefin selectivity increases with the increase of nitrogen content. Alkali metals are typical electronic promoters for FTS catalysts.<sup>27</sup> Qiao and co-workers used potassium-promoted graphene-supported iron catalysts for FTS.<sup>7</sup> The presence of K effectively improves the production of lower olefins. Moreover, in these works,<sup>7,25</sup> the FTS reactions are conducted at 340 °C and H<sub>2</sub>/CO = 1, which are believed to favor the formation of olefins. In this study, however, the FTS reaction is performed at 280 °C and H<sub>2</sub>/CO = 2. Therefore, the low olefin selectivity is obtained on present graphene oxide-supported iron catalysts without any electronic promoters or dopants. Similar results are also reported for other iron/carbon catalysts without promoters.<sup>28</sup> All catalysts are found to have a CO<sub>2</sub> selectivity of 20–27%. It is conceivable that the properties of graphene oxide used in this study resulted in the high WGS activity, due to numerous oxygen-containing groups existing in GO making water access easier.

Fig. 6 presents the CO conversion as a function of time on stream (h) on argon pretreated catalysts. The activity of Fe/GO-300–450 slightly increases with time on stream and then levels off in the range of 14–20%. Meanwhile the Fe/GO-500 catalyst has the highest initial activity, which increases rapidly to a maximum of 39% and then approaches a steady state for 70 h. It is evident that the argon pretreatment at 500 °C largely improves the catalyst activity. In order to further elaborate the effect of argon pretreatment on the catalysts, XRD, Mössbauer and TEM are used to examine the phase composition of the catalysts after 110 h on stream in FTS.

### 3.6 Active phase in FTS

Fig. 7 shows the XRD patterns of the catalysts after the FTS reaction. The predominant iron phase in these used catalysts is Hägg carbide ( $\chi$ -Fe<sub>5</sub>C<sub>2</sub> JCPDS 51-0997), which is considered as the FTS active phase.<sup>29</sup> For Fe/GO-300, -350, and -400 catalysts, just a minor amount of the Fe<sub>3</sub>O<sub>4</sub> phase is detected. While with the increase of calcination temperature, the Fe<sub>3</sub>O<sub>4</sub> peaks become obvious, especially in Fe/GO-450 and Fe/GO-500. This is consistent with previous studies which reported

Table 2 FTS performances of the catalysts

| Catalyst  | CO conversion (%) | FTY ( $\mu\text{mol}_{\text{CO}} \text{g}_{\text{Fe}}^{-1} \text{s}^{-1}$ ) | CO <sub>2</sub> selectivity (mol%) | HC selectivity (wt%) |                                |                 | Olefin/paraffin in C <sub>2</sub> –C <sub>4</sub> |
|-----------|-------------------|-----------------------------------------------------------------------------|------------------------------------|----------------------|--------------------------------|-----------------|---------------------------------------------------|
|           |                   |                                                                             |                                    | CH <sub>4</sub>      | C <sub>2</sub> –C <sub>4</sub> | C <sub>5+</sub> |                                                   |
| Fe/GO     | 3.3               | 30                                                                          | —                                  | —                    | —                              | —               | —                                                 |
| Fe/GO-300 | 16.8              | 109                                                                         | 20.9                               | 29.5                 | 48.5                           | 22              | 0.42                                              |
| Fe/GO-350 | 14.8              | 95                                                                          | 21.3                               | 29.7                 | 48.8                           | 21.5            | 0.43                                              |
| Fe/GO-400 | 17.4              | 109.5                                                                       | 24.5                               | 30.6                 | 49.5                           | 19.9            | 0.42                                              |
| Fe/GO-450 | 22.2              | 137                                                                         | 25.8                               | 25.7                 | 49.7                           | 24.6            | 0.41                                              |
| Fe/GO-500 | 39                | 231.6                                                                       | 27.1                               | 15                   | 47.3                           | 37.7            | 0.44                                              |

Reaction conditions: 200 mg catalyst, H<sub>2</sub>:CO = 2:1, 280 °C, P = 2 MPa, 10 L per (g of catalyst per h). Data are reported at 70 h.



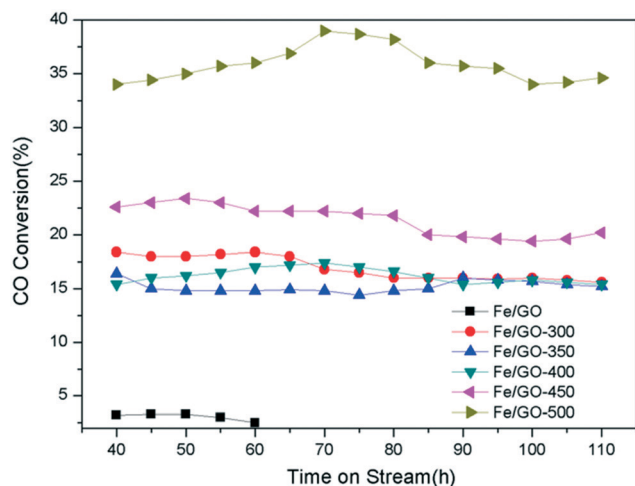


Fig. 6 CO conversion as a function of time on stream (hours) in the FTS reaction at 280 °C.

that significant amounts of  $\text{Fe}_3\text{O}_4$  are found in conventional supported and unsupported catalysts after long-term FTS.<sup>30–32</sup> Clearly, the used Fe/GO-500 catalyst contains the highest amount of magnetite ( $\text{Fe}_3\text{O}_4$  JCPDS 89-4319).

In order to further identify and quantify iron phases in the used catalysts, Mössbauer spectra are collected. Fig. 8 shows the Mössbauer spectra measured at 10 K of the catalysts after the FTS reaction. These spectra are fitted by eight sextets, and the corresponding spectral parameters are given in Table 3. The used catalysts are composed of  $\text{Fe}_3\text{O}_4$ ,  $\chi\text{-Fe}_5\text{C}_2$  and  $\varepsilon\text{-Fe}_2\text{C}$ . Note that the  $\text{Fe}_2\text{C}$  phase is invisible in the XRD patterns. It is probably because the small crystallite sizes of  $\text{Fe}_2\text{C}$  are below the detection limit of XRD. The  $\varepsilon\text{-Fe}_2\text{C}$  phase is commonly reported after carburization at low temperatures for small particle sizes and in the presence of a support.<sup>33</sup> On the basis of this, the appearance of  $\varepsilon\text{-Fe}_2\text{C}$  after reaction in this study is reasonable, which is similar to the observations by de Smit *et al.*<sup>34,35</sup> It is generally acknowledged that

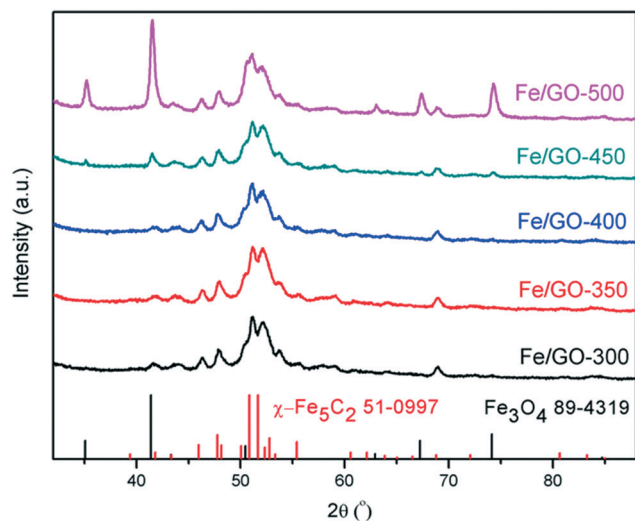


Fig. 7 XRD patterns of the catalysts after FTS.

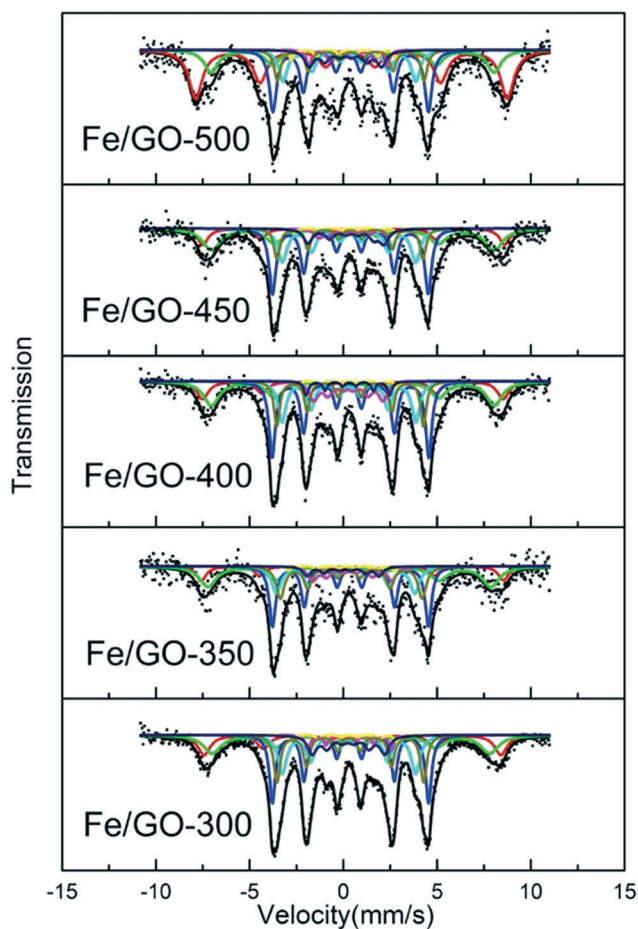


Fig. 8 Mössbauer spectra of the catalysts after FTS at 10 K.

iron carbides are the active phases in FTS. At a typical FTS temperature of 543 K, many studies agree that  $\chi\text{-Fe}_5\text{C}_2$  is the active phase.<sup>34,36</sup> Nevertheless, in low-temperature Fischer-Tropsch synthesis at 423–473 K,  $\varepsilon\text{-Fe}_2\text{C}$  is believed to be more active than  $\chi\text{-Fe}_5\text{C}_2$ , as reported by Qiao and co-workers.<sup>37</sup> The  $\varepsilon\text{-Fe}_2\text{C}$  phase exhibits high selectivity to heavier hydrocarbons and remarkable robustness during the stability test. Later, they further confirm that the activity of iron-boron catalysts evolves positively with the content of  $\varepsilon\text{-Fe}_2\text{C}$  in low-temperature Fischer-Tropsch synthesis at 423 K.<sup>38</sup> Although the catalytic activity of each of these iron carbides with respect to FTS is still controversial, the iron carbides are believed to be the FTS active phases. However, the  $\text{Fe}_3\text{O}_4$  content in the used catalysts increases with increasing calcination temperature, reaching a maximum at Fe/GO-500. Recalling the reaction results, it can be found that the lower the iron carbide content, the higher the FTS activity. This result is inconsistent with the general knowledge that iron carbides are the active phase of the FTS reaction.

The FTS reaction and the accompanying WGS reaction produce  $\text{H}_2\text{O}$  and  $\text{CO}_2$  by-products. The iron carbides can be oxidized by  $\text{H}_2\text{O}$  and  $\text{CO}_2$ .<sup>32</sup> Moreover, it is inevitable that iron carbides can be oxidized by air when the catalysts are removed from the reactor.<sup>39,40</sup> Shroff *et al.* reported that the

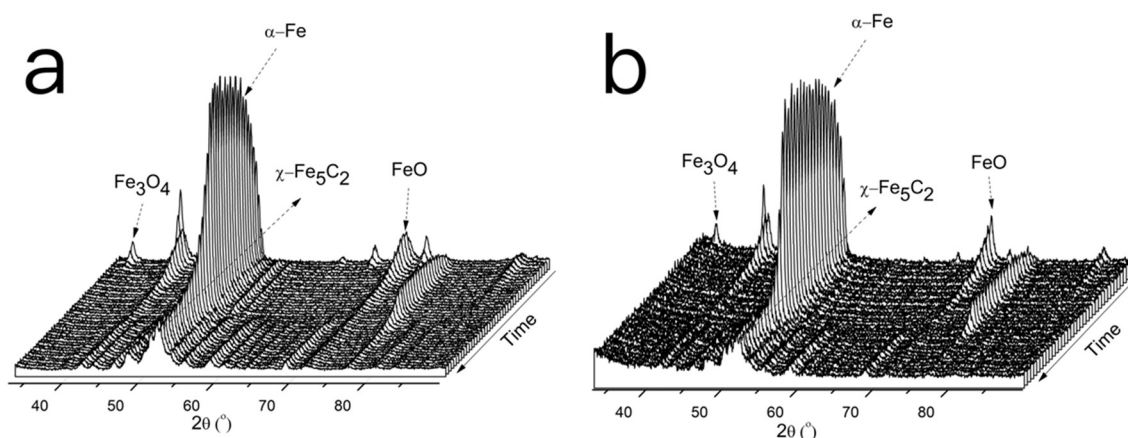
**Table 3** Mössbauer parameters of the catalysts after FTS at 10 K

| Catalysts | IS (mm s <sup>-1</sup> ) | QS (mm s <sup>-1</sup> ) | H <sub>eff</sub> (T) | Phase                              | Area (%) |
|-----------|--------------------------|--------------------------|----------------------|------------------------------------|----------|
| Fe/GO-300 | 0.40                     | 0.16                     | 49.2                 | Fe <sub>3</sub> O <sub>4</sub> (A) | 13.2     |
|           | 0.52                     | -0.22                    | 46.0                 | Fe <sub>3</sub> O <sub>4</sub> (B) | 17.2     |
|           | 0.34                     | 0.07                     | 25.9                 | Fe <sub>5</sub> C <sub>2</sub> (A) | 22.1     |
|           | 0.30                     | -0.01                    | 22.1                 | Fe <sub>5</sub> C <sub>2</sub> (B) | 17.6     |
|           | 0.35                     | -0.04                    | 13.9                 | Fe <sub>5</sub> C <sub>2</sub> (C) | 3.6      |
|           | 0.17                     | -0.29                    | 17.5                 | Fe <sub>2</sub> C(A)               | 1.6      |
|           | 0.33                     | 0.07                     | 24.3                 | Fe <sub>2</sub> C(B)               | 15.9     |
|           | 0.26                     | 0.03                     | 12                   | Fe <sub>2</sub> C(C)               | 8.8      |
|           | 0.34                     | 0.28                     | 50.1                 | Fe <sub>3</sub> O <sub>4</sub> (A) | 9.0      |
| Fe/GO-350 | 0.53                     | -0.41                    | 46.9                 | Fe <sub>3</sub> O <sub>4</sub> (B) | 21.5     |
|           | 0.35                     | 0.05                     | 25.8                 | Fe <sub>5</sub> C <sub>2</sub> (A) | 24.4     |
|           | 0.20                     | -0.16                    | 23.0                 | Fe <sub>5</sub> C <sub>2</sub> (B) | 15.2     |
|           | 0.40                     | 0.15                     | 13.2                 | Fe <sub>5</sub> C <sub>2</sub> (C) | 7.9      |
|           | 0.05                     | -0.30                    | 17.5                 | Fe <sub>2</sub> C(A)               | 1.3      |
|           | 0.36                     | 0.13                     | 23.5                 | Fe <sub>2</sub> C(B)               | 16.1     |
|           | 0.03                     | -0.11                    | 12                   | Fe <sub>2</sub> C(C)               | 4.6      |
|           | 0.40                     | 0.21                     | 49.8                 | Fe <sub>3</sub> O <sub>4</sub> (A) | 11.0     |
|           | 0.61                     | -0.21                    | 46.9                 | Fe <sub>3</sub> O <sub>4</sub> (B) | 20.0     |
| Fe/GO-400 | 0.34                     | 0.08                     | 26.0                 | Fe <sub>5</sub> C <sub>2</sub> (A) | 25.7     |
|           | 0.31                     | -0.03                    | 22.2                 | Fe <sub>5</sub> C <sub>2</sub> (B) | 17.0     |
|           | 0.25                     | 0.02                     | 11.5                 | Fe <sub>5</sub> C <sub>2</sub> (C) | 8.6      |
|           | 0.29                     | -0.12                    | 16.7                 | Fe <sub>2</sub> C(A)               | 1.3      |
|           | 0.35                     | 0.02                     | 24.3                 | Fe <sub>2</sub> C(B)               | 13.2     |
|           | 0.28                     | -0.06                    | 13.9                 | Fe <sub>2</sub> C(C)               | 3.3      |
|           | 0.40                     | 0.31                     | 50.0                 | Fe <sub>3</sub> O <sub>4</sub> (A) | 10.1     |
|           | 0.60                     | -0.20                    | 46.8                 | Fe <sub>3</sub> O <sub>4</sub> (B) | 22.2     |
|           | 0.33                     | 0.08                     | 25.8                 | Fe <sub>5</sub> C <sub>2</sub> (A) | 16.4     |
| Fe/GO-450 | 0.34                     | -0.07                    | 22.1                 | Fe <sub>5</sub> C <sub>2</sub> (B) | 17.6     |
|           | 0.40                     | 0.30                     | 11.6                 | Fe <sub>5</sub> C <sub>2</sub> (C) | 4.32     |
|           | 0.22                     | -0.19                    | 17.1                 | Fe <sub>2</sub> C(A)               | 1.27     |
|           | 0.35                     | -0.01                    | 24.2                 | Fe <sub>2</sub> C(B)               | 10.6     |
|           | 0.29                     | -0.30                    | 12.6                 | Fe <sub>2</sub> C(C)               | 7.6      |
|           | 0.40                     | 0.06                     | 51.7                 | Fe <sub>3</sub> O <sub>4</sub> (A) | 30.4     |
|           | 0.54                     | -0.10                    | 46.9                 | Fe <sub>3</sub> O <sub>4</sub> (B) | 14.6     |
|           | 0.32                     | 0.12                     | 25.8                 | Fe <sub>5</sub> C <sub>2</sub> (A) | 21.0     |
|           | 0.35                     | -0.04                    | 22.0                 | Fe <sub>5</sub> C <sub>2</sub> (B) | 12.3     |
| Fe/GO-500 | 0.35                     | -0.22                    | 13.0                 | Fe <sub>5</sub> C <sub>2</sub> (C) | 3.6      |
|           | 0.12                     | -0.15                    | 17.5                 | Fe <sub>2</sub> C(A)               | 2.2      |
|           | 0.39                     | -0.01                    | 24.3                 | Fe <sub>2</sub> C(B)               | 9.1      |
|           | 0.12                     | -0.03                    | 12.0                 | Fe <sub>2</sub> C(C)               | 6.7      |

catalyst surface consists almost entirely of magnetite with some iron carbide embedded in the core, when the sample is removed from the reactor without careful passivation.<sup>40</sup>

In order to reveal the origin of the Fe<sub>3</sub>O<sub>4</sub> phase in the used catalysts, *in situ* XRD is utilized to trace the iron phase transformation of catalysts under a FTS environment. The catalysts are *in situ* reduced in hydrogen at 300 °C and then exposed in syngas at 280 °C. Fig. 9 shows the XRD patterns of Fe/GO-300 and Fe/GO-500 during FTS. As we can see, the initial phase of Fe<sub>2</sub>O<sub>3</sub> is quickly reduced to Fe<sub>3</sub>O<sub>4</sub> and then transforms into an FeO and  $\alpha$ -Fe mixed phase. When all the iron oxides are completely reduced to  $\alpha$ -Fe, the temperature is cooled down to 280 °C and the gas is changed to syngas. The  $\alpha$ -Fe phase gradually disappears, and iron carbide phases come into existence. During the reaction process, there is no Fe<sub>3</sub>O<sub>4</sub> detected in both catalysts. Thus, the large content of Fe<sub>3</sub>O<sub>4</sub> in Fe/GO-500 (as shown in the XRD and MES results) after reaction is ascribed to the re-oxidation of iron carbides, when the catalysts are withdrawn from the reactor after reaction. The Fe/GO-300 catalyst with GO coverage has poor capability for gas diffusion. Thus, a small amount of iron carbides is re-oxidized into Fe<sub>3</sub>O<sub>4</sub> after the reaction. While for the Fe/GO-500 sample, the partially destroyed GO structure with more defects makes more iron phases exposed to the environment and leads to a higher Fe<sub>3</sub>O<sub>4</sub> content after unloading from the reactor. This indirectly confirms that there are more exposed catalytic sites in the Fe/GO-500 catalyst, which is consistent with its higher FTS activity.

The TEM images of Fe/GO-500 after 110 h on stream in FTS are shown in Fig. 10. It is observed that the iron NPs are well dispersed on GO without aggregation and some of them are not covered with GO sheets. The average size of Fe/GO-300, -350, -400, -450 and -500 after FTS is about 21, 24, 24.6, 28.5 and 33.1 nm, respectively (ESI†). The decrease of crystallite sizes after reaction is ascribed to the breakage of large iron NPs. It is reported that the iron particle sizes above 6 nm almost have no effect on the production of lower olefins from syngas.<sup>41</sup> It is justified that the crystallite size of the catalysts in this study is not the main reason for the differences in FTS performances. The relatively unchanged high dispersion of NPs after 110 h FTS are attributed to the presence of oxygen-containing groups and defects on GO sheets. They act

**Fig. 9** *In situ* XRD patterns of (a) Fe/GO-300; (b) Fe/GO-500.



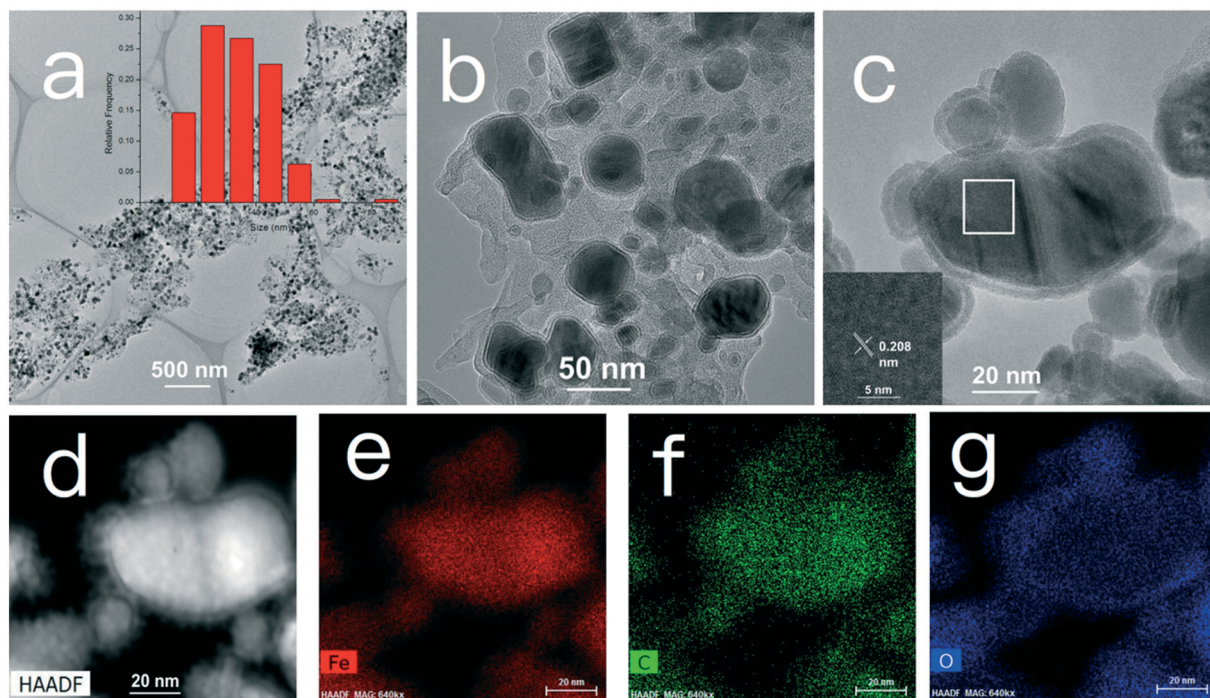


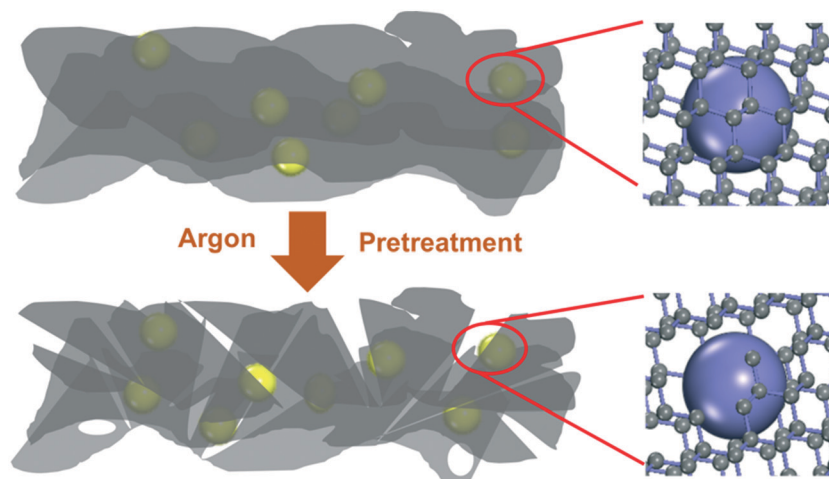
Fig. 10 (a)–(c) TEM images, (d) STEM images and the corresponding elemental mapping of (e) Fe, (f) C, and (g) O of the Fe/GO-500 catalyst after FTS.

as anchoring sites and inhibit the aggregation of NPs.<sup>5,7</sup> Additionally, the graphene sheets have numerous wrinkles and folds, providing more reactive sites. Further information about the iron NPs can be obtained from the zoom-in TEM images (Fig. 10b and c). They clearly present a single-crystalline core–amorphous shell structure. The Hägg carbide particles are covered with a thin layer of amorphous  $\text{Fe}_3\text{O}_4$  due to the re-oxidation once exposed in air. Moreover, the observed void between the core and shell is due to the density mismatch for oxide and carbide phases, also reported by Li *et al.*<sup>31</sup>

The inset of Fig. 10c exhibits a fringe spacing of about 0.208 nm corresponding to the value of the (021) plane of the

$\chi\text{-Fe}_5\text{C}_2$  phase (JCPDS 51-0997) with a single crystalline feature. In addition, the STEM image (Fig. 10d) and the corresponding elemental mapping images present obvious Fe, C and O distributions in Fe/GO-500 after FTS. The STEM image exhibits a characteristic light/dark contrast with changes in focus, which is typical of voids. The darker green (C) in the core and the darker blue (O) in the shell demonstrate the  $\chi\text{-Fe}_5\text{C}_2$  core and  $\text{Fe}_3\text{O}_4$  shell structure. Moreover, some small particles without shells are assigned to the  $\text{Fe}_3\text{O}_4$  phase, which may be due to the complete oxidation of some iron carbide particles with small sizes.

The GO support has a stacked layered structure with some oxygen-containing groups and defects, which are beneficial



Scheme 1 Schematic illustration of the argon pretreatment on catalysts.



for anchoring iron NPs to obtain a high dispersion. With the increase of argon pretreatment temperature, the NPs tend to sinter in bigger size (as seen from the XRD and TEM results), leading to the weaker interaction between iron oxide NPs and GO. This, in turn, promotes the reduction of catalysts in H<sub>2</sub>/CO. Moreover, the membrane-like GO structure is destroyed and more defects are formed under high temperature argon pretreatment, as indicated by the BET, TEM and Raman results. Thus, more catalytically active sites are provided. Considering the discussion above, it is believed that the argon pretreatment does affect the FTS performances of catalysts. To the best of our knowledge, it is the first time to reveal the argon pretreatment as a successful method to enhance the Fischer–Tropsch performances of graphene oxide-supported iron catalysts (Scheme 1).

## 4. Conclusions

In summary, a graphene oxide-supported iron oxide NP composite is synthesized and used as a FTS catalyst. The effects of different argon pretreatment temperatures on the textural, morphological and surface properties and the FTS performances of graphene oxide-supported iron oxide NPs are investigated. The pretreatment of catalysts in argon at 500 °C for 5 h can efficiently enhance the surface area and defects and promote the reduction and carbonization behaviors. The Fe–GO interaction is weakened, and more active sites are exposed for FTS, thus high FTS activity and C<sub>5+</sub> selectivity are achieved. The argon pretreatments of samples pave a new road for the development of GO-supported FTS catalysts. More novel methods are needed to fabricate efficient GO-supported iron catalysts with excellent FTS performances.

## Conflicts of interest

There are no conflicts to declare.

## Acknowledgements

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## References

- Q. Zhang, J. Kang and Y. Wang, *ChemCatChem*, 2010, 2, 1030–1058.
- H. M. T. Galvis, J. H. Bitter, C. B. Khare, M. Ruitenbeek, A. I. Dugulan and K. P. de Jong, *Science*, 2012, 335, 835–838.
- J. M. Martinmartinez and M. A. Vannice, *Ind. Eng. Chem. Res.*, 1991, 30, 2263–2275.
- W. Chen, Z. Fan, X. Pan and X. Bao, *J. Am. Chem. Soc.*, 2008, 130, 9414–9419.
- S. O. Moussa, L. S. Panchakarla, M. Q. Ho and M. S. El-Shall, *ACS Catal.*, 2014, 4, 535–545.
- H. Zhao, Q. Zhu, Y. Gao, P. Zhai and D. Ma, *Appl. Catal., A*, 2013, 456, 233–239.
- Y. Cheng, J. Lin, K. Xu, H. Wang, X. Yao, Y. Pei, S. Yan, M. Qiao and B. Zong, *ACS Catal.*, 2016, 6, 389–399.
- Y. Wei, R. Ding, C. Zhang, B. Lv, Y. Wang, C. Chen, X. Wang, J. Xu, Y. Yang and Y. Li, *J. Colloid Interface Sci.*, 2017, 504, 593.
- E. S. Lox and G. F. Froment, *Ind. Eng. Chem. Res.*, 1993, 32, 71–82.
- C. H. Zhang, Y. Yang, B. T. Teng, T. Z. Li, H. Y. Zheng, H. W. Xiang and Y. W. Li, *J. Catal.*, 2006, 237, 405–415.
- A. Ganguly, S. Sharma, P. Papakonstantinou and J. Hamilton, *J. Phys. Chem. C*, 2011, 115, 17009–17019.
- Z. Ji, G. Zhu, X. Shen, H. Zhou, C. Wu and M. Wang, *New J. Chem.*, 2012, 36, 1774–1780.
- J. Su, M. Cao, L. Ren and C. Hu, *J. Phys. Chem. C*, 2011, 115, 14469–14477.
- Z. X. Huanwen Wang, H. Yi, H. Wei, Z. Guo and X. Wang, *Nano Energy*, 2014, 7, 86–96.
- D. Fu, W. Dai, X. Xu, W. Mao, J. Su, Z. Zhang, B. Shi, J. Smith, P. Li, J. Xu and Y.-F. Han, *ChemCatChem*, 2015, 7, 752–756.
- R. Zboril, M. Mashlan and D. Petridis, *Chem. Mater.*, 2002, 14, 969–982.
- S. Yang, X. Song, P. Zhang and L. Gao, *ACS Appl. Mater. Interfaces*, 2015, 7, 75–79.
- W. Chen, X. Duan, G. Qian, D. Chen and X. Zhou, *ChemSusChem*, 2015, 8, 2927–2931.
- S. Kundu, Y. Wang, W. Xia and M. Muhler, *J. Phys. Chem. C*, 2008, 112, 16869–16878.
- A. Y. Khodakov, A. Griboval-Constant, R. Bechara and V. L. Zholobenko, *J. Catal.*, 2002, 206, 230–241.
- S. Taghavi, A. Asghari and A. Tavasoli, *Chem. Eng. Res. Des.*, 2017, 119, 198–208.
- W. Chen, X. L. Pan, M. G. Willinger, D. S. Su and X. H. Bao, *J. Am. Chem. Soc.*, 2006, 128, 3136–3137.
- W. Chen, X. Pan and X. Bao, *J. Am. Chem. Soc.*, 2007, 129, 7421–7426.
- Y. Yang, H. W. Xiang, Y. Y. Xu, L. Bai and Y. W. Li, *Appl. Catal., A*, 2004, 266, 181–194.
- X. Chen, D. Deng, X. Pan, Y. Hu and X. Bao, *Chem. Commun.*, 2015, 51, 217–220.
- P. Zhai, C. Xu, R. Gao, X. Liu, M. Li, W. Li, X. Fu, C. Jia, J. Xie and M. Zhao, *Angew. Chem., Int. Ed.*, 2016, 55, 9902–9907.
- J. Li, X. Cheng, C. Zhang, J. Wang, W. Dong, Y. Yang and Y. Li, *J. Chem. Technol. Biotechnol.*, 2017, 92, 1472–1480.
- M. Oschatz, S. Krause, N. A. Krans, C. Hernández Mejía, S. Kaskel and K. P. de Jong, *Chem. Commun.*, 2017, 53, 10204–10207.
- J. Xu and C. H. Bartholomew, *J. Phys. Chem. B*, 2005, 109, 2392–2403.
- T. Riedel, H. Schulz, G. Schaub, K. W. Jun, J. S. Hwang and K. W. Lee, *Top. Catal.*, 2003, 26, 41–54.
- S. Li, G. D. Meitzner and E. Iglesia, *J. Phys. Chem. B*, 2001, 105, 5743–5750.
- A. Sarkar, D. Seth, A. K. Dozier, J. K. Neathery, H. H. Hamdeh and B. H. Davis, *Catal. Lett.*, 2007, 117, 1–17.

- 33 D. B. Bukur, K. Okabe, M. P. Rosynek, C. P. Li, D. J. Wang, K. Rao and G. P. Huffman, *J. Catal.*, 1995, **155**, 353–365.
- 34 E. de Smit, A. M. Beale, S. Nikitenko and B. M. Weckhuysen, *J. Catal.*, 2009, **262**, 244–256.
- 35 E. de Smit, F. Cinquini, A. M. Beale, O. V. Safonova, W. van Beek, P. Sautet and B. M. Weckhuysen, *J. Am. Chem. Soc.*, 2010, **132**, 14928–14941.
- 36 T. Herranz, S. Rojas, F. J. Perez-Alonso, M. Ojeda, P. Terreros and J. L. G. Fierro, *J. Catal.*, 2006, **243**, 199–211.
- 37 K. Xu, B. Sun, J. Lin, W. Wen, Y. Pei, S. Yan, M. Qiao, X. Zhang and B. Zong, *Nat. Commun.*, 2014, **5**, 5783.
- 38 K. Xu, Y. Cheng, J. Lin, H. Wang, S. Xie, Y. Pei, S. Yan, M. Qiao, Z. H. Li and B. Zong, *J. Catal.*, 2016, **339**, 102–110.
- 39 J. W. Niemantsverdriet, A. M. V. D. Kraan, W. L. V. Dijk and H. S. V. D. Baan, *J. Phys. Chem.*, 1980, **84**, 3363–3370.
- 40 M. D. Shroff and A. K. Datye, *Catal. Lett.*, 1996, **37**, 101–106.
- 41 H. M. T. Galvis, J. H. Bitter, T. Davidian, M. Ruitenbeek, A. I. Dugulan and K. P. de Jong, *J. Am. Chem. Soc.*, 2012, **134**, 16207–16215.